

Urinary Tract Infection and Bacteriuria in Pregnancy



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KEYWORDS

- Urinary tract infection • Bacteriuria • Antimicrobials • Urinary tract infection • Pyelonephritis
- Pregnancy

KEY POINTS

- Antimicrobial choice in pregnancy should reflect safety for both the mother and the fetus.
- During pregnancy, asymptomatic bacteriuria (ASB) significantly increases the risk of pyelonephritis and subsequent maternal and fetal complications. Treatment of ASB significantly reduces these risks.
- Pregnant patients should be screened for ASB at least once early in pregnancy, and, if culture is positive, treated for 3 to 7 days. These patients should be followed with serial cultures throughout pregnancy, and prophylactic antimicrobial therapy should be considered.
- Pregnant patients with cystitis should also be treated for 3 to 7 days. These patients should be followed with serial cultures throughout pregnancy, and prophylactic antimicrobial therapy should be considered.
- Pregnant patients with pyelonephritis should initially be admitted for intravenous antimicrobial therapy and receive a total of 7 to 14 days of culture-directed treatment. These patients should be followed with serial cultures throughout pregnancy, and prophylactic antimicrobial therapy should be strongly considered.

INTRODUCTION

Urinary tract infections (UTIs) during pregnancy may be classified as asymptomatic bacteriuria (ASB), infections of the lower urinary tract (cystitis), or infections of the upper urinary tract (pyelonephritis). Lower tract bacteriuria (ASB or cystitis) is associated with a 20% to 30% increased risk of developing pyelonephritis in pregnancy, likely

due to the physiologic changes of the urinary tract during pregnancy.^{1–3} Both lower and upper urinary tract bacteriuria, including ASB, are associated with adverse maternal and fetal outcomes, including preterm birth.^{4–10} Therefore, screening and treatment of bacteriuria during pregnancy have become standard of care recommended by multiple professional organizations.^{2,11–15}

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PATHOGENESIS

The urinary tract is sterile under normal circumstances. Bacteriuria generally occurs because of ascending bacteria from a fecal reservoir or vaginal/perineal skin flora.¹⁶ Pathogenic organisms in pregnant women are similar to those found in the nonpregnant population. *Escherichia coli* is the most common pathogen, representing 70% to 80% of all UTIs in pregnancy.^{7,8,16–19} Enterobacteriaceae, including *Klebsiella* and *Enterobacter*, are also common pathogens, and infections with other gram-negative organisms such as *Proteus*, *Pseudomonas*, and *Citrobacter* can occur.^{7,8,16–20} Gram-positive organisms, mostly group B *Streptococci* (GBS), are also common pathogens and are the cause of up to 10% of UTIs in pregnant women.^{17,20} Infection with other microorganisms, including *Mycoplasma hominis*, *Ureaplasma parvum*, *Gardnerella vaginalis*, lactobacilli, and *Chlamydia trachomatis*, have also been described.^{21–23}

URINARY TRACT CHANGES IN PREGNANCY

The entire urinary tract is anatomically and physiologically altered in pregnant women (Table 1).¹ Together, these upper and lower urinary tract changes are thought to increase the risk of development of pyelonephritis from ASB and lower UTIs.^{4,7} During pregnancy, renal length increases by approximately 1 cm and glomerular filtration rate increases by approximately 30% to 50%.¹ Mild hydroureteronephrosis is observable as early as the seventh week of gestation.⁵ This dilation is due to the overall decreased peristalsis in the collecting system and ureters, attributable to both the muscle-relaxing effects of progesterone and the progressive mechanical obstruction from the gravid uterus.^{1,3,5,24} During pregnancy, the

bladder also experiences progressive superior and anterior displacement as well as hypertrophy and smooth muscle relaxation, leading to increased capacity and urinary stasis.^{1,3,24–26}

ANTIMICROBIALS IN PREGNANCY

Choice of Treatment

Choice of antimicrobials during pregnancy must reflect safety considerations for both the mother and the fetus. In the mother, physiologic changes in pregnancy can alter pharmacokinetics and decrease serum drug concentrations; changes include increased maternal fluid intravascular and extravascular fluid volumes, increased renal blood flow, increased glomerular filtration rate, and drug distribution to the fetus.^{27,28} Most antimicrobials cross the placenta, and drugs that can cause teratogenesis or other harm to the fetus should be avoided. Despite this known risk, few large-scale trials on antimicrobial use during pregnancy have been performed because of logistical and ethical issues. Much of the known drug safety data derives from studies when pregnant women were not excluded from new therapies,^{29,30} or from data extracted from animal models and observational studies.^{27,28,31}

The US Food and Drug Administration (FDA) previously published a classification system for drugs in pregnancy, labeling medications either class A, B, C, D, or X. This system was widely panned as being unhelpful because of a lack of high-quality data, therefore leading to classification of most drugs, including antimicrobials, as category C (“animal studies have shown an adverse effect on the fetus and there are no adequate and well-controlled studies in humans, but potential benefits may warrant use of the drug in pregnant women despite potential risks”).^{28,32} In December 2014, the FDA approved a rule to remove the pregnancy categories A, B, C, D, and X from drug product labeling and instead instituted a new system for labeling medications in pregnancy that is intended to be more practical and help guide decision-making by physicians and patients.³³ Labeling changes go into effect in June 2015.

Despite a lack of high-quality safety data, many antimicrobials have been used in pregnancy for years without adverse maternal or fetal outcomes, including penicillins, cephalosporins, clindamycin, and macrolides (Table 2).^{27,28} Recent meta-analyses found no evidence to support one particular antimicrobial over another for either ASB³⁴ or symptomatic UTIs.³⁵ Therefore, after reviewing safety profiles of antimicrobials, choice of empiric treatment should reflect local antibiograms and

Table 1 Urinary tract changes in pregnancy	
Kidneys	Increased renal length Increased glomerular filtration rate by 30%–50%
Collecting System	Decreased peristalsis
Ureters	Decreased peristalsis Mechanical obstruction
Bladder	Displaced anteriorly and superiorly Smooth muscle relaxation Increased capacity

Data from Waltzer WC. The urinary tract in pregnancy. J Urol 1981;125(3):271–6.

Table 2
Antimicrobials in pregnancy

Drug	Fetal Toxicity	Comments	Prior FDA Category
Penicillins			
Penicillin G	Low risk	Commonly used	B
Amoxicillin	Low risk	Commonly used	B
Ampicillin	Low risk	Commonly used	B
Cephalosporins			
Cephalexin	Low risk	1st generation; commonly used	B
Cefuroxime	Low risk	2nd generation; commonly used	B
Ceftriaxone	Low risk	3rd generation; commonly used	B
Ceftazidime	Low risk (limited data)	3rd generation with antipseudomonal coverage	B
Cefepime	Low risk (limited data)	4th generation with antipseudomonal coverage	B
Monobactams			
Aztreonam	Low risk (limited data)	Limited data; consider ID consultation	B
Carbapenems			
Imipenem	Very limited data	Limited data; must be administered with cilastatin; consider ID consultation	C
Meropenem	Very limited data	Limited data; consider ID consultation	B
Lincosamides			
Clindamycin	Low risk	Commonly used; useful in penicillin-allergic patients	B
Macrolides			
Azithromycin	Low risk	Commonly used	B
Erythromycin	Low risk	More gastrointestinal side effects than azithromycin	
Nitrofurans			
Nitrofurantoin	Controversial teratogenic risk; risk of hemolytic anemia in G6PD deficiency during third trimester	Commonly used; use for only for lower UTIs	B
Phosphonics			
Fosfomycin	Low risk (limited data)	Use for only for lower UTIs	B
Sulfonamides			
Sulfadiazine	Risk of antifolate teratogenesis; risk of hyperbilirubinemia in third trimester	Avoid if alternatives available	C
Trimethoprim-sulfamethoxazole	Risk of antifolate teratogenesis	Avoid in pregnancy	C
Glycopeptides			
Vancomycin	Very limited data	Limited data; consider ID consultation	B

(continued on next page)

Table 2 (continued)			
Drug	Fetal Toxicity	Comments	Prior FDA Category
Oxazolidinones			
Linezolid	Only case reports available	Limited data; consider ID consultation	C
Lipopeptides			
Daptomycin	Only case reports available	Limited data; consider ID consultation	B
Aminoglycosides			
Gentamycin	Potential risks of ototoxicity and nephrotoxicity	Commonly used; useful for pyelonephritis	C
Fluoroquinolones			
Ciprofloxacin	Theoretic risk of arthropathy	Avoid in pregnancy	C
Tetracyclines			
Tetracycline	Risk of teratogenicity and discoloration of teeth/bones	Avoid in pregnancy	D
Doxycycline	—	Avoid in pregnancy	D

Abbreviations: G6PD, glucose-6-phosphate dehydrogenase; ID, infectious disease.
Data from Refs.^{18,27,28,44}

susceptibility patterns, and final treatment should be tailored to culture-directed data.¹¹ Cost-effective choices should also be considered.³⁶ Finally, as with all patients, health care providers should prescribe antimicrobials for pregnant patients only for appropriate indications and duration, and after discussing the benefits and potential risks of treatment.

Antimicrobial Resistance

The prevalence of multi-drug-resistant (MDR) organisms is increasing at an alarming pace in the general population.^{37–40} In addition, few drugs are being developed to combat the spread of resistance.^{37,39} Although the spread of MDR organisms to the pregnant population has not been widely reported, it is likely that this challenge will be encountered in the future,⁴¹ stressing the need for culture-directed therapy and clinician-driven antimicrobial stewardship.⁴²

Penicillins

Penicillins are one of the oldest groups of antimicrobials and have been used for many years in pregnancy without known adverse effects.^{43,44} Their safety profile has been confirmed in multiple studies.^{31,44–47} These β -lactams include penicillin G, ampicillin, and amoxicillin. Although resistance is common, these drugs are considered first-line therapy for susceptible bacteria, especially for GBS.^{28,31,36,43,48}

Cephalosporins

Cephalosporins are also a group of β -lactam antimicrobials with a well-established safety profile in pregnancy.^{27,28,31,36,43,44,48,49} Susceptible bacteria vary with the different generations. Third-generation cephalosporins are particularly useful because of their excellent coverage of gram-negative organisms as well as some gram-positive organisms and are often used empirically in the treatment of pyelonephritis. In addition, certain third-generation cephalosporins such as ceftazidime have antipseudomonal activity. It is important to remember that none of the cephalosporins are effective against Enterococcus. More data in pregnancy are available for the earlier generations of cephalosporins, including cephalexin and cefuroxime, yet overall the cephalosporins are considered generally safe in pregnancy.^{28,48}

Monobactams

Aztreonam is the only monocyclic β -lactam antimicrobial currently commercially available. It is effective against many gram-negative organisms and has low cross-reactivity in penicillin-allergic patients, but must be administered intravenously or intramuscularly. Limited data are available on the use of aztreonam in pregnancy, yet the risk of adverse events or teratogenicity is considered low.^{27,50}

Carbapenems

Carbapenems are a class of β -lactam antimicrobials that are resistant to the extended-spectrum β -lactamase enzymes made by MDR gram-negative bacteria.^{40,51} They are available in intravenous form only. Because of their limited indications for use and relatively recent development, data on safety of these drugs in pregnancy are scarce.^{27,36}

Clindamycin

Clindamycin, a lincosamide antimicrobial, is useful in penicillin-allergic patients for treatment of infections caused by *Staphylococcus* and GBS. It is not active against *Enterococcus* or most aerobic gram-negative bacteria. Clindamycin can cause diarrhea in up to 10% of patients, and rarely, pseudomembranous colitis, but there are no known studies associating clindamycin with teratogenicity.^{27,28,36,44,52}

Macrolides

The macrolide antimicrobials, such as erythromycin, azithromycin, and clarithromycin, are useful for treatment of *Staphylococcus* infections, with limited activity against gram-negative pathogens. The most common side effect is gastrointestinal upset, especially with erythromycin. Macrolide antimicrobials have a long safety record in pregnancy, and large studies have not found associations between macrolide use and teratogenic risk.^{27,28,36,53,54}

Nitrofurantoin

Nitrofurantoin is a unique antimicrobial specifically useful for lower UTIs. It achieves therapeutic levels in the urine but not in tissue and therefore should not be used to treat pyelonephritis. For lower UTIs, nitrofurantoin is useful because of limited resistance and bactericidal activity against many common uropathogens, including *E coli*, *Enterococcus faecalis*, *Klebsiella* spp, *Staphylococcus saprophyticus*, and *Staphylococcus aureus*, although it does not have activity against *Proteus*, *Serratia*, or *Pseudomonas*.^{36,55}

Safety data for nitrofurantoin are conflicting. The National Birth Defects Prevention Study recently demonstrated an association between nitrofurantoin and congenital malformations, such as anophthalmia, microphthalmia, atrial septal defects, and cleft lip and palate.³¹ However, results from this retrospective case-control study should be interpreted with caution because of recall bias and other flaws associated with this study design (for example, only 35% of patients could recall the drug name they were given).^{31,48} Two recent

population-based studies demonstrated no increase in teratogenicity in patients exposed to nitrofurantoin.^{56,57} The American Congress of Obstetricians and Gynecologists (ACOG) supports appropriate use of nitrofurantoin in the first trimester when no suitable alternatives are available, and as a first-line agent in the second and third trimesters.⁴⁸

Aside from potential teratogenic risks, there are known rare but serious risks, including pulmonary toxicity⁵⁸ and hemolytic anemia, in patients with glucose-6-phosphate dehydrogenase deficiency; nitrofurantoin should therefore be avoided in these patients.⁵⁹ Nitrofurantoin is also contraindicated at term and in the neonate less than 1 month of age because neonates have immature enzymatic activity and insufficient quantities of reduced glutathione in circulating neonatal red blood cells, leading to risk of hemolytic anemia.^{60,61}

Fosfomycin

Fosfomycin, a phosphonic acid derivative, is a broad-spectrum antimicrobial not commonly used in clinical practice in the United States that may see resurgence in use because of increasing antimicrobial resistance.^{62,63} It has no structural analogues to other antimicrobials and is useful against many MDR organisms, including extended-spectrum β -lactamase-producing organisms, *Pseudomonas*, methicillin-resistant *S aureus*, and vancomycin-resistant *Enterococcus*.^{62,63} Use of fosfomycin in pregnancy has been studied primarily because it achieves high concentrations in the urine for up to 3 days, making it an attractive choice for treatment of uncomplicated lower UTIs.^{62–64} Gastrointestinal upset is the most common adverse event.⁶³ Although there are no adequate studies specifically examining teratogenic risk, several trials have examined the use of single-dose fosfomycin for treatment of ASB and lower UTIs in pregnancy with no reported adverse outcomes.^{34,62,65–68} Overall, fosfomycin has a broad spectrum of activity against many MDR organisms, and it may have an increasing role in the antimicrobial armamentarium for treatment of lower UTIs.

Sulfonamides and Trimethoprim

Sulfonamides and the combination drug trimethoprim-sulfamethoxazole are not recommended as first-line agents in pregnancy. Sulfonamides inhibit folate synthesis and therefore have been implicated in antifolate teratogenicity.^{27,28,36,69} In the National Birth Defects Prevention Study, sulfonamides were associated with

more congenital defects than any other antimicrobial.³¹ However, other studies have not demonstrated this risk,⁷⁰ and ACOG states that sulfonamide treatment remains appropriate in the first trimester when no other suitable antimicrobial alternatives are available.⁴⁸ Aside from teratogenic risk, sulfonamides have also been reported to cause hyperbilirubinemia and kernicterus when used in the third trimester.²⁷

Trimethoprim, a dihydrofolate reductase inhibitor, is also associated with antifolate teratogenicity and should be avoided in pregnancy.⁶⁹

Aminoglycosides

The aminoglycosides, gentamycin, amikacin, and tobramycin, are effective for treating gram-negative bacilli and are often useful in the treatment of pyelonephritis in combination with ampicillin.^{28,70} However, aminoglycosides have potential risks of ototoxicity and nephrotoxicity in both the mother and the fetus.^{27,28,71,72} Gentamycin is the preferred aminoglycoside in obstetrics, if needed, because of its comparatively superior safety profile and low cost.³⁶ Despite potential risks, no teratogenicity or neonatal ototoxicity or nephrotoxicity has been reported after administration of gentamycin in pregnancy, and gentamycin is widely used.^{28,44}

Quinolones

Quinolones, including ciprofloxacin and levofloxacin, achieve high tissue concentrations, making them effective choices for pyelonephritis in nonpregnant patients, yet they are generally avoided during pregnancy because of possible arthropathies and teratogenicity.^{27,28} Quinolone exposure in animal models can induce abnormal cartilage development.⁷³ Interestingly, despite this risk, when pregnant women were exposed to quinolones, adverse events have not been consistently demonstrated.^{31,74,75} Overall, this class of antimicrobials should be avoided in pregnancy if suitable alternatives are available.

Tetracyclines

Tetracycline and doxycycline should not be used in pregnancy, because they are associated with teratogenicity as well as discoloration of the bones and teeth.^{27–30}

Vancomycin

Vancomycin is a glycopeptide antimicrobial useful for treatment of methicillin-resistant *S aureus* and other resistant gram-positive organisms such as *Enterococcus*. Its use is associated with potential

nephrotoxicity and ototoxicity,⁷⁶ although these effects have not been demonstrated in the few published reports of vancomycin use in pregnancy.^{44,77,78} Because of the very limited evidence of vancomycin use in pregnancy, it should be used with caution only when no alternatives exist.

Linezolid, Daptomycin, and Newer Antimicrobials

Even less is known about the use of newer antimicrobials in pregnancy at this time. Only case reports of linezolid, an oxazolidinone antimicrobial,^{79,80} and daptomycin, a lipopeptide antimicrobial,⁸¹ have been published in the literature to date. Fittingly, these drugs should be used with extreme caution in pregnancy and with consideration of infectious disease consultation.

ASYMPTOMATIC BACTERIURIA IN PREGNANCY

Definition

ASB is the presence of significant bacteriuria without symptoms, such as dysuria, urgency, frequency, hematuria, or suprapubic discomfort. The Infectious Diseases Society of America (IDSA) defines ASB as 2 consecutive voided cultures with 10⁵ or more colony-forming units of the same bacteria, or a single catheterized urine specimen with 10² or more colony-forming units of one bacterium.² This definition is based on observations from asymptomatic nonpregnant women who, when screened for ASB with urine cultures, only had confirmed bacteriuria in a second culture 80% of the time.^{4,82–84} The United States Preventative Services Task Force (USPTF), ACOG, and others recommend screening with one urine culture and do not clarify whether this culture should be repeated to confirm the diagnosis of ASB.^{12–15} Although a single culture may overestimate the prevalence of bacteriuria, it is reasonable to treat ASB without a confirmatory culture in the pregnant population.

Epidemiology

Pregnancy alone is not a risk factor for ASB; ASB occurs in 2% to 10% of pregnant and nonpregnant women (see **Table 3**).^{2,85–87} The incidence of ASB in nonpregnant women increases with advancing age, the presence of diabetes, sexual activity, history of prior UTI, and anatomic or functional urinary tract abnormalities.^{2,85,88–90} In pregnant women, socioeconomic status and parity have been inconsistently associated with prevalence of ASB.^{91–95}

Table 3
Incidence, diagnosis, and recommended treatment and follow-up of asymptomatic bacteriuria, cystitis, and pyelonephritis

	Incidence (%)	Diagnosis	Treatment Duration (d)	Follow-up
ASB	2–10	• No symptoms • Bacteriuria	3–7	• Periodic screening for recurrent bacteriuria • Consider antimicrobial prophylaxis
Lower UTI (cystitis)	1–2	• Dysuria • Urgency • Frequency • Hematuria • Suprapubic discomfort • Bacteriuria	3–7	• Periodic screening for recurrent bacteriuria • Consider antimicrobial prophylaxis
Upper UTI (pyelonephritis)	≤1	• Fever • Chills • Flank pain • Nausea • Vomiting • Bacteriuria	7–14	• Periodic screening for recurrent bacteriuria • Strongly consider antimicrobial prophylaxis

Data from Refs. ^{2,12,34,35,124}

Significance

Unlike in the general population, ASB should be screened for and treated in pregnant women. If left untreated, up to 40% of pregnant women with ASB may develop pyelonephritis.^{2,4–6} Treatment of ASB with antimicrobials significantly decreases the risk of symptomatic infection and pyelonephritis.^{6,96,97} Untreated bacteriuria is also associated with preterm birth and low birth weight, likely due to association with pyelonephritis,^{20,21,96,98–102} although this link remains controversial due to possible confounders and has not been consistently reported.^{103–105}

Because of the outdated nature of much of the data linking ASB to pyelonephritis from the early antibiotic era, unclear methodology in these publications, and the possible consequences of antimicrobial overuse in the era of MDR organisms, a study is currently re-examining the significance of ASB. This trial is examining the benefits and cost-effectiveness of screening low-risk pregnant women for ASB at 16 and 22 weeks' gestation and then randomizing patients with a positive culture to treatment with either placebo or 5 days of nitrofurantoin (Netherlands Trial Registry NTR3068).¹⁰⁶ Primary outcomes are maternal pyelonephritis and preterm delivery. This study is being performing in the Netherlands, one of the few countries where screening and treatment of ASB in pregnancy is not currently standard of care.

Screening

The USPTF, IDSA, ACOG, American Academy of Pediatrics (AAP), and the American Academy of Family Physicians (AAFP) recommend screening pregnant women for ASB.^{2,12,14,15} The IDSA, AAP, and ACOG recommend pregnant women be screened with a urine culture early in pregnancy,^{2,14} whereas the USPTF and AAFP recommend screening with a urine culture at 12–16 weeks' gestation, or at the first prenatal visit, if this occurs later.^{12,15}

If the initial culture is negative, the IDSA and others do not make a specific recommendation for or against repeated screening for ASB, because there is limited evidence to guide timing of screening.² Conflicting data exist on the incidence of ASB after 20 weeks' gestation.^{87,107,108} However, only 1% to 2% of women with an initial negative culture obtained early in pregnancy will develop pyelonephritis later, and no data exist to date that additional screening decreases this risk.^{2,84,86,109}

The gold standard screening test for ASB in pregnancy is a urine culture.^{2,12} Culture should be obtained midstream.¹¹⁰ Many studies have examined alternative tests because urine cultures are expensive, are labor-intensive, and require a period of 24 to 48 hours before a result is determined; however, at this time, an inexpensive, rapid, sensitive, and specific alternative to culture has not been proven. Urinalysis and urine dipstick are specific, but not sensitive, for bacteriuria, and

therefore, use in screening would result in many false-negative results.^{13,107,111–118} Urine dipslide, a variant method of urine culture that uses an agar-soaked plastic paddle attached to a plastic vial, is both sensitive and specific for bacteriuria in pregnancy and could be useful in countries with limited resources and microbiologist availability.¹¹⁴ However, further validation is needed, especially in standard daily practice.¹¹⁹ Therefore, in light of the consequences of bacteriuria and pyelonephritis in the obstetric population, and because culture data are vital to tailor appropriate antimicrobial therapy in the era of increased antimicrobial resistance, urine culture remains the recommended screening test.

Treatment

Antimicrobial treatment of ASB in pregnancy eliminates bacteriuria and significantly reduces the incidence of pyelonephritis.⁹⁶ There is insufficient evidence to recommend one specific drug over another,³⁴ and the ideal treatment duration also remains debatable, ranging from a single dose to 7 days of treatment. Most single-dose regimens are based on the unique pharmacokinetic properties of fosfomycin, which achieves high concentrations in the urine for up to 3 days after treatment with a single 3 g oral dose.^{63,64,66–68} However, single doses of nitrofurantoin, sulfa, amoxicillin, and trimethoprim have also been examined.^{34,67} A *Cochrane Review* of the duration of antimicrobial therapy in 13 studies reported that single-dose therapy may be less effective than a 7-day regimen.³⁴ The IDSA recommends a treatment duration of 3 to 7 days.²

Follow-up and Prevention

Options for follow-up after treatment of ASB include close surveillance or antimicrobial prophylactic therapy. Up to one-third of patients treated for ASB will have recurrent bacteriuria after treatment if prophylaxis is not instituted, and therefore, periodic screening for recurrent bacteriuria is recommended.^{2,120,121} Daily prophylactic therapy with 50 to 100 mg of nitrofurantoin should also be considered. Although there is limited high-quality evidence that antimicrobial prophylaxis decreases bacteriuria in pregnancy,^{122,123} this is a well-established therapy in nonpregnant patients.¹²⁴ Postcoital prophylaxis can also be considered.¹²⁵ However, nonpharmacologic prophylaxis such as cranberry juice supplementation is not recommended, because it has not been proven to reduce ASB in pregnancy and is associated with a high dropout rate, and efficacy even in nonpregnant patients remains controversial.¹²⁶

LOWER URINARY TRACT INFECTIONS (CYSTITIS)

A symptomatic lower UTI (cystitis) is diagnosed based on the presence of symptoms such as dysuria, urgency, frequency, hematuria, and suprapubic discomfort in combination with bacteriuria. The incidence of cystitis during pregnancy is less well characterized than ASB but is estimated at 1% to 2% (**Table 3**).^{98,101,127}

Diagnosis, treatment, and follow-up of symptomatic lower UTI are identical to those for ASB. Urine culture remains the diagnostic modality of choice. As with ASB, there is no evidence to support one specific antimicrobial regimen over another for treatment of symptomatic UTI, and treatment is recommended for 3 to 7 days.³⁵ Follow-up, including periodic screening for recurrent bacteriuria and consideration of daily antimicrobial prophylactic therapy, is also recommended after treatment.^{122–125}

UPPER URINARY TRACT INFECTIONS (PYELONEPHRITIS)

Epidemiology

The incidence of pyelonephritis in the ASB screening era is 1% or less (see **Table 3**).^{17,20,99,128} This incidence is decreased from an incidence of 4% or higher before routine screening and treatment of ASB was initiated.^{4–6} Risk factors for pyelonephritis include ASB, young age, nulliparity, previous episodes of pyelonephritis, sickle-cell disease/trait, diabetes, and other immunosuppression.^{17,20,128} Pyelonephritis occurs more often in the second and third trimesters, when stasis and hydronephrosis are most evident, and only 10% to 20% of pyelonephritis occur in the first trimester.^{17,129}

Significance

As stated previously, up to 40% of pregnant women with untreated ASB develop pyelonephritis,^{2,4–6} which may also be associated with preterm birth and low birth weight,^{20,21,96,98–102} although this remains controversial.^{103–105} In addition to the potential detrimental effects on the fetus, pyelonephritis is also associated with maternal morbidity. Maternal complications of pyelonephritis include anemia, acute kidney injury, sepsis, pulmonary insufficiency, and acute respiratory distress syndrome.^{17,20,98,128–130} Maternal morbidity of pyelonephritis is similar across all 3 trimesters.¹²⁹

Diagnosis

Pyelonephritis is a clinical diagnosis consisting of symptoms such as fever, chills, flank pain, nausea,

vomiting, and costovertebral angle tenderness in the presence of bacteriuria and pyuria.^{11,16,131,132} Imaging is not required to diagnose pyelonephritis. Urine culture is the diagnostic test of choice. Whether routine blood cultures should also be obtained for pregnant patients with pyelonephritis has not yet been adequately studied. Approximately 20% to 30% of blood cultures will be positive in pregnant patients with pyelonephritis.^{17,133} However, discrepancies between positive blood and urine cultures are rare and excluding routine blood cultures would result in significant cost savings.¹³³ A recent *Cochrane Review* concluded that there was insufficient evidence at this time to conclude if routine blood cultures should be obtained in pregnant women with pyelonephritis.¹³⁴

Treatment

Pregnant patients with pyelonephritis should be admitted and treated with intravenous antimicrobials and hydration. Initial antimicrobial therapy is empiric and should be tailored to culture results. Empiric treatment options include ampicillin plus gentamycin or a single-agent cephalosporin such as ceftriaxone.^{35,132,133} Historically, these regimens have resulted in 95% or better response rates within 72 hours,^{17,35,120,133,135,136} although resistance to ampicillin alone approaches 50% in contemporary studies.¹³³ No single-treatment regimen has been proven to be superior to others,^{17,35,133,135,136} and individual treatments should be tailored to local antibiograms and susceptibility patterns.

There are no organizational guidelines on the duration of treatment of pyelonephritis in pregnant women; however, most recommend a treatment course of 7 to 14 days,^{16,17,35,132,133} in line with IDSA recommendations for nonpregnant patients.¹¹ After treatment, follow-up cultures should be obtained, as with ASB and cystitis. Prophylactic antimicrobial therapy should be strongly considered to reduce the risk of recurrent pyelonephritis.^{122–125,137} In a study of 400 pregnant patients with pyelonephritis, 2.7% of women were readmitted for recurrent pyelonephritis, all of whom were noncompliant with antimicrobial prophylaxis, reinforcing the importance of this therapy.¹⁷

In patients who do not respond to culture-directed treatment, alternative diagnoses should be considered and an imaging study such as a renal ultrasound should be obtained to rule out renal abscess, nephrolithiasis, or other structural abnormality, although it is important to remember that some degree of hydronephrosis is expected in pregnancy.^{1,16,132} In a recent study of 105

pregnant women with pyelonephritis, 45% had right hydronephrosis, 17% had left hydronephrosis, and only 14% had a normal ultrasound, yet all patients were managed with antimicrobials alone.⁹⁹

SUMMARY

Bacteriuria is a common complication of pregnancy that may lead to significant adverse maternal and fetal outcomes. Antimicrobial selection should reflect safety considerations for both the mother and the fetus. ASB, cystitis, and pyelonephritis should be treated promptly, and patients should be followed closely after treatment, with strong consideration of daily antimicrobial prophylaxis.

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